Dylan R. Zelko

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EDUCATION

Master of Science - Robotics Expected Dec 2026
University at Buffalo, School of Engineering and Applied Sciences

Bachelor of Science - Mechanical Engineering; Minor in Robotics Sep 2021 - May 2025
University at Buffalo, School of Engineering and Applied Sciences

WORK EXPERIENCE

Robotics Design Researcher May 2026 - Present
Applied Medical Coatings / Human-in-the-Loop Systems Lab, Buffalo, NY

- Redesign a magnetically actuated variable-stiffness robotic arm carriage to enable precise manipulation and adjustable compliance in industrial pick-and-place applications
- Develop mechanical assemblies in Fusion 360, integrating compact belt-and-pulley systems, bearing supports, and enclosed components to improve structural rigidity, reliability, and manufacturability
- Manufacture and assemble custom 3D-printed components for robotic arm integration and functional testing

Additive Manufacturing Technician Aug 2025 - Present
Digital Manufacturing Lab, University at Buffalo, Buffalo, NY

- Fabricate and manage 3D printing work orders using FDM, SLA, PolyJet, and CFF technologies, ensuring efficient turnaround times with high-quality prints
- Provide technical support and guidance to students, faculty, and industry on 3D printing, including material selection, print orientation, support strategies, and post-processing techniques

PROJECTS

NASA Lunabotics, University at Buffalo Sep 2024 - Present
Founder & Project Lead, Buffalo, NY Link to Demo

- Founded and led a multidisciplinary team developing a lunar excavation rover, coordinating mechanical, electrical, and software engineering efforts
- Spearheaded mechanical design of custom traction wheels, a modular chassis, and a rotating drum excavator; conducted structural FEA using ANSYS and applied DEM simulations to improve rover and excavation design
- Earned Best First-Year Presentation and Demonstration and the Phoenix Award for Most Efficient Battery Usage at the NASA Lunabotics Competition

Space Bulls (University Rover Challenge), University at Buffalo Apr 2023 - May 2025
Founder & Project Lead, Buffalo, NY Link to Demo

- Founded and led a multidisciplinary team developing a Mars rover for extreme terrain, overseeing subsystem integration, engineering documentation, and project milestones
- Designed custom rover chassis, suspension, and mechanical subsystems in Fusion 360 for durability, modularity, and off-road performance

SKILLS

Technical Skills: Fusion 360, SolidWorks, ANSYS, Microsoft Suite (Excel, Teams, PowerPoint), 3D printing (FDM, SLA, PolyJet, CFF) including printer maintenance and material selection, machining
Programming: MATLAB, Python, ROS2, C++

AWARDS, LEADERSHIP & ADDITIONAL EXPERIENCE

- NASA New York Space Grant Internship Award, 2025
- Barbara McDaniel Undergraduate Scholarship by the National Training and Simulation Association, 2024
- President, UB Robotics Club, Sep 2022 - May 2025
- Publication: K. Wang, D. Zelko, M. Delano, *Textile band electrodes as an alternative to spot Ag/AgCl electrodes for calf bioimpedance measurements*, Biomedical Physics & Engineering Express, 6(1), 015010, 2020. doi:10.1088/2057-1976/ab5b02
- Graduate Teaching Assistant, MAE 451/494, University at Buffalo, Aug 2025 - Present